

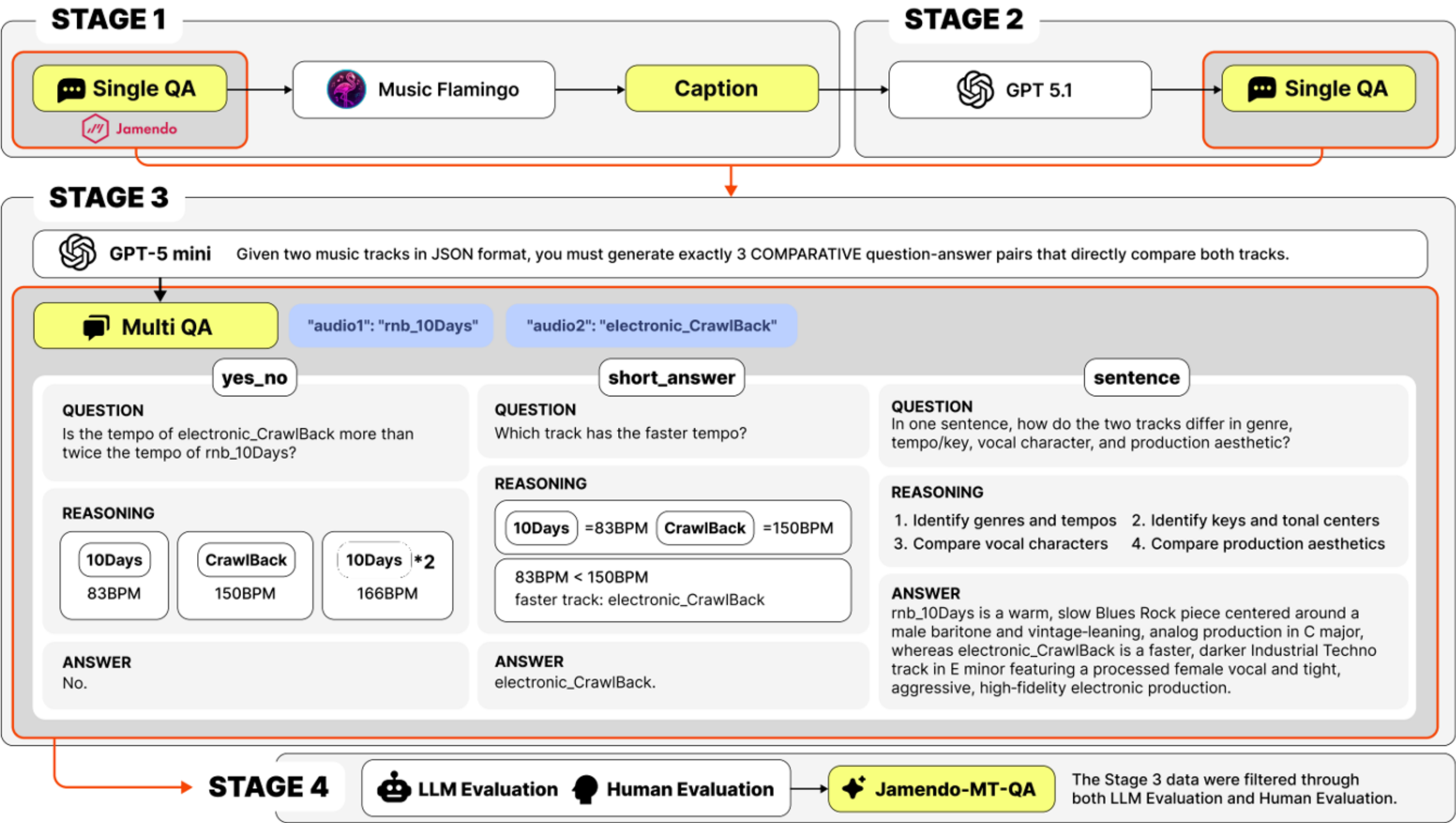
Jamendo-MT-QA : A Benchmark for Multi-Track Comparative Music Question Answering

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Jamendo-MT-QA : Dataset Construction Pipeline



Motivation

Most existing music datasets focus on single-track understanding. However, human music perception is often comparative: we describe one track as faster, darker, more energetic, or more vocal-driven than another. Existing Music QA datasets such as Jamendo-QA [1] provide useful foundations for single-track question answering, but they do not directly test whether models can compare multiple pieces of music and explain their differences.

We present **Jamendo-MT-QA**, a benchmark for **multi-track comparative music question answering**, designed to evaluate whether audio-language models can reason over relationships between multiple songs rather than understanding each track in isolation. Built on **Jamendo-QA** [1] with a **Music Flamingo** [2] captioning backbone, our pipeline first extracts rich musical descriptions for individual tracks and then generates comparative QA pairs across track pairs.

Human Evaluation Validity

Metric	R1	R2	R3	R4	Human Mean	GPT-5 Mini Mean	MAD
Correctness	4.93	4.66	4.64	5.00	4.79	4.87	0.32
Comparative Validity	4.93	4.75	4.83	4.99	4.83	4.61	0.44
Reasoning Quality	4.87	4.71	4.67	5.00	4.78	4.37	0.56
Difficulty	1.89	2.58	3.12	2.66	2.25	2.17	0.57

Reference

[1] Junyoung Koh, Soo Yong Kim, Yongwon Choi, and Gyu Hyeong Choi. **Jamendo-QA : A Large-Scale Music Question Answering Dataset**. ArXiv. 2025.

[2] Sreyan Ghosh, Arushi Goel, Lasha Koroshinadze, Sang-gil Lee, Zhifeng Kong, Joao Felipe Santos, Ramani Duraiswami, Dinesh Manocha, Wei Ping, Mohammad Shoeybi, Bryan Catanzaro. **Music Flamingo: Scaling Music Understanding in Audio Language Models**. International Conference on Machine Learning, 2026.

Jamendo-MT-QA Example

GPT-5 mini

yes no

Question: Is pop_WhentheSnowfalls slower in tempo than pop_LonesomeRoad?

Reasoning :

Step 1: Identify tempos for both tracks: pop_WhentheSnowfalls ≈ 81 BPM, pop_LonesomeRoad = 125 BPM.

Step 2: Compare numerical BPM values: 81 is less than 125.

Step 3: Conclude whether the first track is slower than the second based on the BPM comparison.

Answer : Yes

short answer

Question: Which track features a warm, finger-picked acoustic guitar carrying the core chordal rhythm, as opposed to being built primarily around synth pads and electronic drums?

Reasoning :

Step 1: Examine instrumentation of pop_WhentheSnowfalls: primarily expansive synth pads, synth bass, and a clean electronic drum kit.

Step 2: Examine instrumentation of pop_LonesomeRoad: warm, finger-picked acoustic guitar carries the core chordal rhythm, with layered synth pads as accompaniment.

Step 3: Determine which audio_name matches the description of a finger-picked acoustic guitar as the core chordal element.

Answer : pop_LonesomeRoad

pop_LonesomeRoad **pop_WhentheSnowfalls**

sentence

Question: In one sentence, how do the two tracks differ in key, tempo, primary instrumentation, and vocal timbre?

Reasoning :

Step 1 : Note key and tempo for pop_WhentheSnowfalls (C major, ≈81 BPM) and pop_LonesomeRoad (E major, 125 BPM).

Step 2 : Compare primary instrumentation: Snowfalls centers on reverberant synth pads, deep synth bass and electronic drums; LonesomeRoad centers on finger-picked acoustic guitar, melodic electric bass, brushed drums and supporting synth pads.

Step 3 : Compare vocal timbres: Snowfalls features a smooth, slightly breathy male baritone with heavy reverb/delay, while LonesomeRoad features a clear, slightly breathy male tenor with lighter reverb and compression.

Step 4 : Synthesize these points into a single comparative sentence.

Answer : Pop_WhentheSnowfalls is rooted in C major at ≈81 BPM and emphasizes lush synth pads, deep synth bass, electronic drums and a smooth, slightly breathy male baritone drenched in reverb/delay, whereas pop_LonesomeRoad sits in E major at 125 BPM and foregrounds a warm finger-picked acoustic guitar, melodic electric bass and brushed drums with layered synth ambience and a clear, slightly breathy tenor delivered with lighter spatial effects.

Jamendo-MT-QA Benchmark

SINGLE AUDIO		MULTI AUDIO									
Music Flamingo, Mu-LLaMA		GPT-4o mini, Qwen2-Audio, Qwen3-Omni									
Model	Type	Yes/No Acc.	Short Acc.	BLEU	R-1	R-2	R-L	BERT-F1	GPT	Claude	
Music Flamingo	Cap	82.1%	88.1%	2.06	25.3	5.9	19.6	0.875	3.25	3.97	
Qwen2-Audio	Cap	26.9%	55.5%	0.58	15.3	1.6	12.1	0.845	1.37	1.60	
MU-LLaMA	Cap	23.6%	50.5%	0.68	16.1	1.9	13.3	0.853	1.92	2.13	
Qwen2-Audio	Multi	34.4%	77.7%	0.57	14.3	1.4	11.3	0.845	1.63	1.75	
Qwen3-Omni	Multi	59.7%	75.5%	1.22	19.9	3.9	16.0	0.860	2.57	3.25	
GPT-4o-mini-Audio	Multi	77.3%	73.2%	0.87	17.8	2.7	14.7	0.859	2.92	3.14	
GPT-4o-Audio	Multi	69.6%	84.4%	1.04	20.0	3.6	16.7	0.870	3.17	3.57	

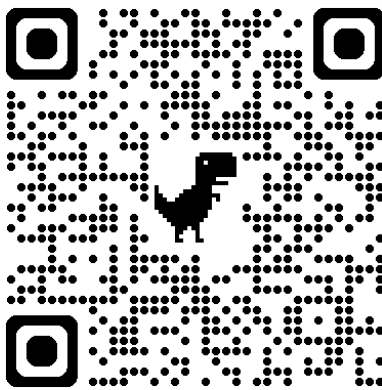
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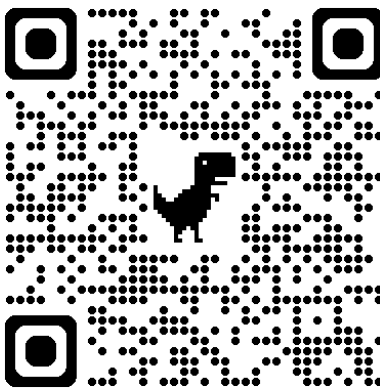
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Jamendo-QA Dataset

